

Divyanshu Pabia

Systems Engineer · AI Systems at Scale



Summary

I am focused on understanding and improving the systems that run modern AI models, from **GPU kernels** and memory behavior to compilers, **inference runtimes**, and **distributed execution**. I enjoy reading systems code, profiling bottlenecks, and building experiments that turn performance questions into measurable results. My background spans backend engineering, iOS development, and network-monitoring software for Tor, and I am now specializing in AI systems, GPU computing, and **efficient LLM inference**.

Experience

The Tor Project

Aug 2025 – July 2026
Remote Contract

Network Health Team

- Contributed to reimplementing the **Java Tor Metrics Library** in **Rust** to retrieve, parse, and validate network descriptors from CollecTor, focusing on memory safety and extensible parsing.
- Worked on a unified descriptor pipeline using object storage, a dedicated parser service, and **PostgreSQL** and **VictoriaMetrics** exports to support historical and real-time network analysis.



Infosys

Mar 2024 – Feb 2025
Mysuru, Karnataka

iOS and backend development intern

- Built a Hospital Management System with a Swift iOS client and **Java/Spring Boot REST APIs** for staff management, bookings, and real-time updates. Implemented **JWT authentication** and role-based access.
- Containerized backend services with **Docker** and supported test/staging CI/CD. Strengthened API design, OpenAPI documentation, and verification through JUnit 5 and integration tests.



iOS App Development Centre

Mar 2023 – Mar 2024
Chennai, Tamil Nadu

Apple and Infosys iOS App Development Program

- Built Gamergrid through the Apple and Infosys program, integrating esports events, scores, and news into a Swift iOS app. Developed practical skills in **URLSession**, **Combine**, REST API integration, pagination, and resilient data fetching.



Tech Analyst

Feb 2022 – July 2022
Chennai, Tamil Nadu

Software development intern

- Built an expense-management backend with **Spring Boot and Hibernate**, using **Kafka and RabbitMQ** for asynchronous communication. Gained hands-on experience in transaction processing, microservices, and event-driven design.

Education

B.Tech SRM Institute of Science and Technology, Computer Science and Engineering
CGPA: 8.5/10.

Chennai, India
Sept 2021 – June 2025

Projects

GeoGPT-RAG

[GitHub](#)

Developed a geospatial assistant prototype combining **retrieval over reference material**, language-model interaction, and GIS tooling. Explored how to connect natural-language requests with spatial data and tools while keeping analysis steps inspectable.

Linux virtual LIFO device

[GitHub](#)

Implemented a virtual last-in, first-out character device as a loadable **Linux kernel module**. Worked with module and device interfaces, device-state management, and the user/kernel boundary to understand how applications interact with kernel-backed services.

ARM microarchitecture in VHDL

[GitHub](#)

Built a **VHDL microarchitecture** that executes a subset of ARM instructions, connecting instruction decoding, datapath behavior, and control logic. Developed a practical understanding of how instruction semantics map to hardware execution.

Gamergrid

[GitHub](#)

Built a native Swift app for following esports events, live scores, match statistics, and personalized news. Integrated **third-party REST APIs** using URLSession and Combine, with attention to pagination, resilient networking, and a responsive mobile interface.

Publications

A Decentralized Blockchain Framework for Secure, Transparent and Privacy-Preserving Toll Payment Systems

July 2025

Divyanshu Pabia, Manasvi Rao Kanukolan, A. Anbarasi

[10.1109/INCSST64791.2025.11210370](#) (2025 International Conference on Smart & Sustainable Technology (INCSST))

Skills

Programming and systems: C/C++, Python, Java, Rust, Linux, Git, CMake

ML systems: PyTorch, JAX, Transformers, torch.compile, vLLM

GPU and performance: CUDA C++, Triton, parallel computing, profiling, Nsight Systems / Compute

Backend and cloud: Spring Boot, PostgreSQL, REST APIs, AWS, Docker, Kubernetes, Terraform

iOS development: Swift, Xcode, URLSession, Combine

Mathematics: Linear algebra, probability, calculus, optimization